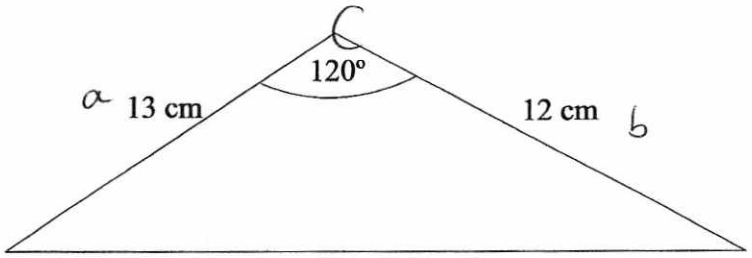
1 Area $= \frac{1}{2} ab \sin C$



Work out the area of the triangle.
Give your answer to 1 decimal place.

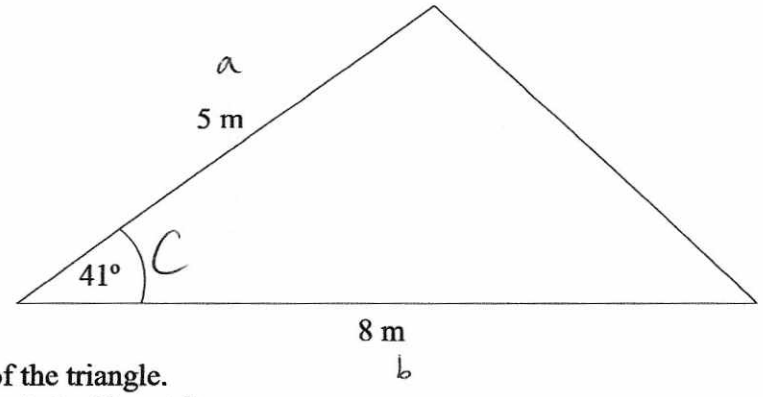
$\frac{1}{2}(13)(12)\sin(120)$

$= 67.5$ (1dp)

67.5 ..... cm^2

(Total for question 1 is 2 marks)

2



Work out the area of the triangle.
Give your answer to 3 significant figures.

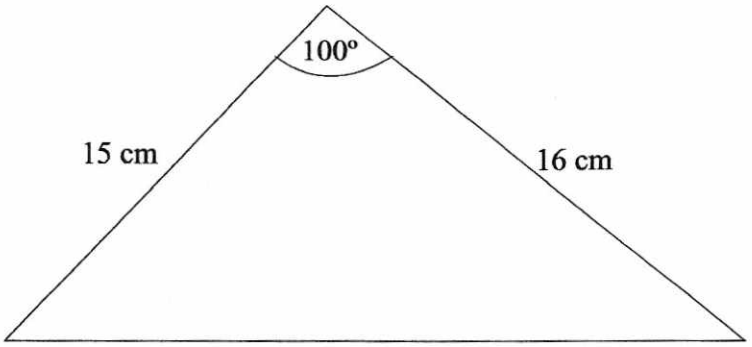
$\frac{1}{2}(5)(8)\sin(41)$

$= 13.1$ 3sf

13.1 ..... m^2

(Total for question 2 is 2 marks)

3



Work out the area of the triangle.
Give your answer to 3 significant figures.

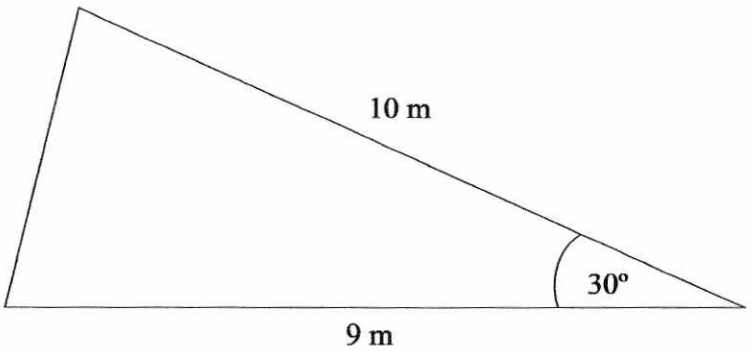
$\frac{1}{2}(15)(16)\sin(100)$

$= 118$ (3sf)

118 ..... cm^2

(Total for question 3 is 2 marks)

4



Work out the area of the triangle.

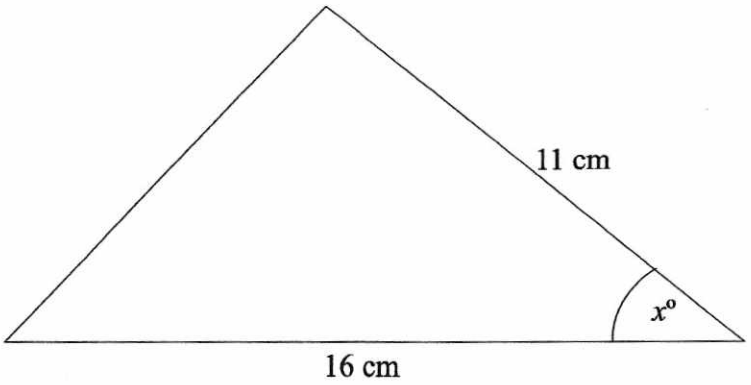
$\frac{1}{2}(10)(9)\sin(30)$

$= 22.5$

22.5 ..... m^2

(Total for question 4 is 2 marks)

5



The area of the triangle is 70cm^2
Work out the value of x.
Give your answer to 1 decimal place.

$\frac{1}{2}(16)(11)\sin(x) = 70$

$88\sin(x) = 70$

$\sin(x) = \frac{70}{88}$

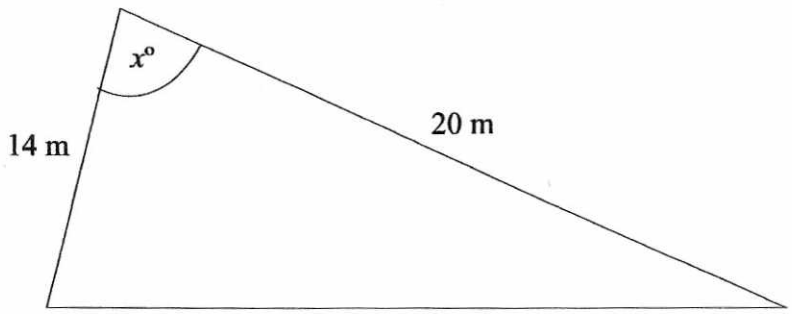
$x = \sin^{-1}\left(\frac{70}{88}\right)$

$= 52.7$ (1dp)

52.7

(Total for question 5 is 3 marks)

6



The area of the triangle is 100m^2
Work out the value of x.
Give your answer to 3 significant figures.

$\frac{1}{2}(14)(20)\sin x = 100$

$140\sin x = 100$

$\sin x = \frac{100}{140}$

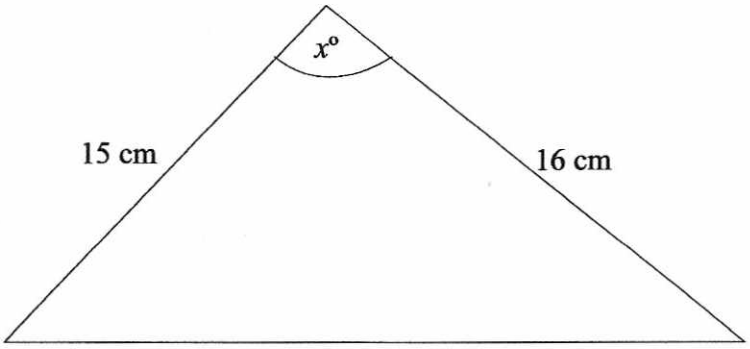
$x = \sin^{-1}\left(\frac{100}{140}\right)$

$= 45.6$ (3sf)

45.6

(Total for question 6 is 3 marks)

7



The area of the triangle is 105cm^2
Work out the value of x.
Give your answer to 1 decimal place.

$\frac{1}{2}(15)(16)\sin x = 105$

$120 \sin x = 105$

$\sin x = \frac{105}{120}$

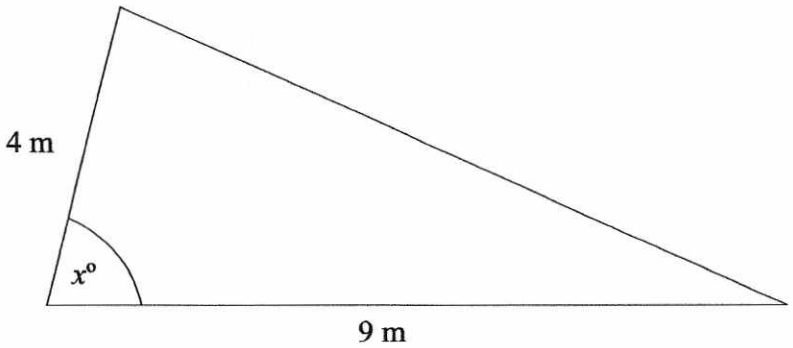
$x = \sin^{-1}\left(\frac{105}{120}\right)$

$= 61.0$ (1dp)

61.0

(Total for question 7 is 4 marks)

8



The area of the triangle is 15m^2
Work out the value of x.
Give your answer to 3 significant figures.

$\frac{1}{2}(4)(9)\sin x = 15$

$18 \sin x = 15$

$\sin x = \frac{15}{18}$

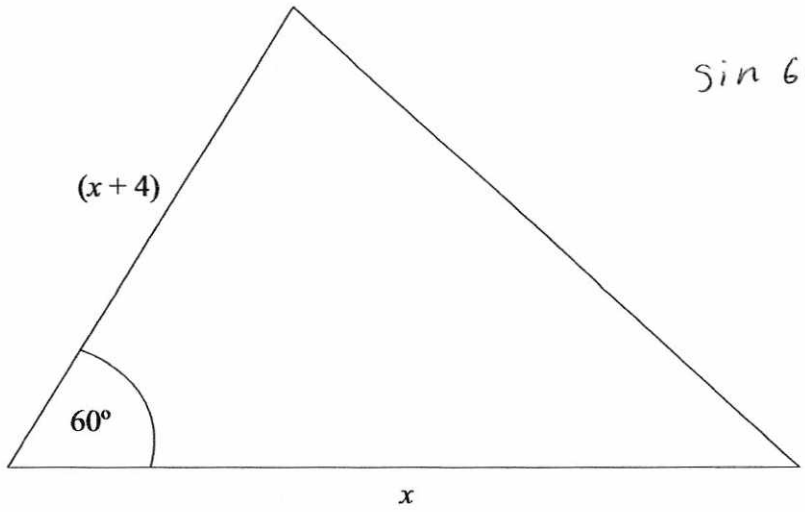
$x = \sin^{-1}\left(\frac{15}{18}\right)$

$= 56.4$ (3sf)

56.4

(Total for question 8 is 4 marks)

9



$\sin 60 = \frac{\sqrt{3}}{2}$

The area of the triangle is $15\sqrt{3}$ cm^2.
Work out the value of x.

$$\frac{1}{2}(x)(x+4)\sin(60) = 15\sqrt{3}$$

$$\frac{1}{2}(x)(x+4)\frac{\sqrt{3}}{2} = 15\sqrt{3}$$

$$\frac{\sqrt{3}}{4}x(x+4) = 15\sqrt{3}$$

$$\sqrt{3}x(x+4) = 60\sqrt{3}$$

$$x(x+4) = 60$$

$$x^2+4x = 60$$

$$x^2+4x-60 = 0$$

$$(x+10)(x-6) = 0$$

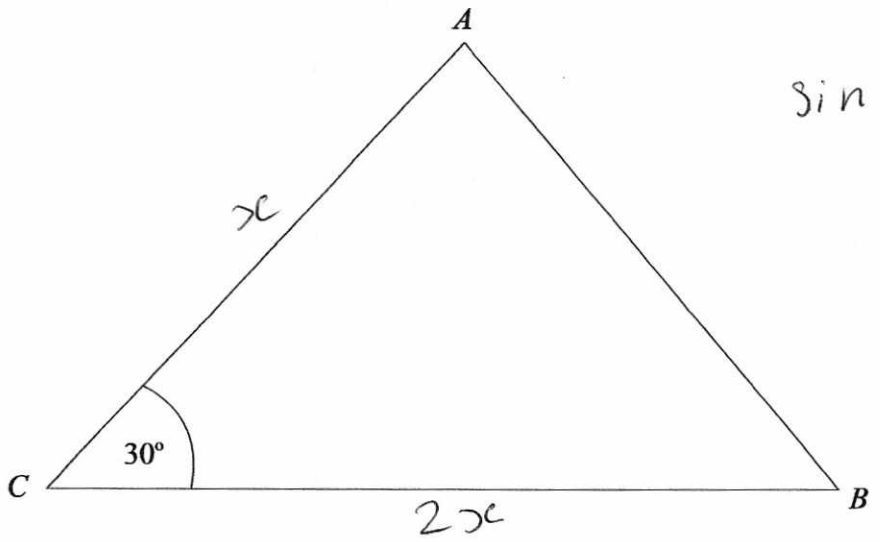
$$x=-10 \quad x=6$$

a length cannot be negative $\therefore$ $\underline{\underline{x=6}}$

6

(Total for question 9 is 4 marks)

10



$\sin 30 = \frac{1}{2}$

The ratio of the length BC to the length AC is 2:1
The area of the triangle is 50cm^2

Work out the value of x.

$$\frac{1}{2}(x)(2x)\sin(30) = 50$$

$$\frac{1}{2}(x)(2x)\left(\frac{1}{2}\right) = 50$$

$$\frac{1}{2}x^2 = 50$$

$$x^2 = 100$$

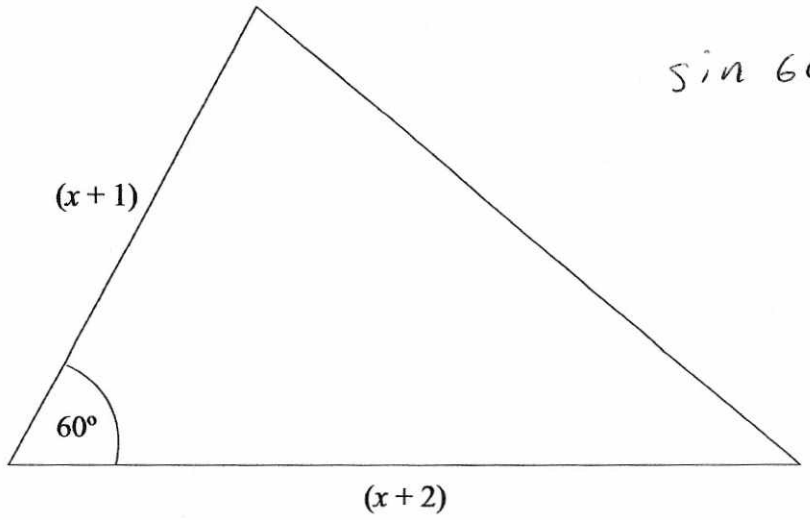
$$x = 10$$

(x cannot be -10 as a length can't be $-$ve)

10

(Total for question 10 is 4 marks)

11



$\sin 60 = \frac{\sqrt{3}}{2}$

The area of the triangle is 25 cm^2.
Work out the value of x.
Give your answer to 3 significant figures.

$$\frac{1}{2}(x+1)(x+2)\sin 60 = 25$$

$$(x+1)(x+2)\left(\frac{\sqrt{3}}{2}\right) = 50$$

$$(x+1)(x+2) = \frac{100}{\sqrt{3}}$$

$$x^2 + 2x + x + 2 = \frac{100}{\sqrt{3}}$$

$$x^2 + 3x + 2 = \frac{100}{\sqrt{3}}$$

$$x^2 + 3x - \text{"}55.735\text{"} = 0$$

$a = 1$
$b = 3$
$c = -55.735\ldots$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$= 6.11$ (3sf) or -9.11 (3sf)

~~x c~~ lengths cannot be negative.

$x = 6.11$

6.11

(Total for question 11 is 5 marks)